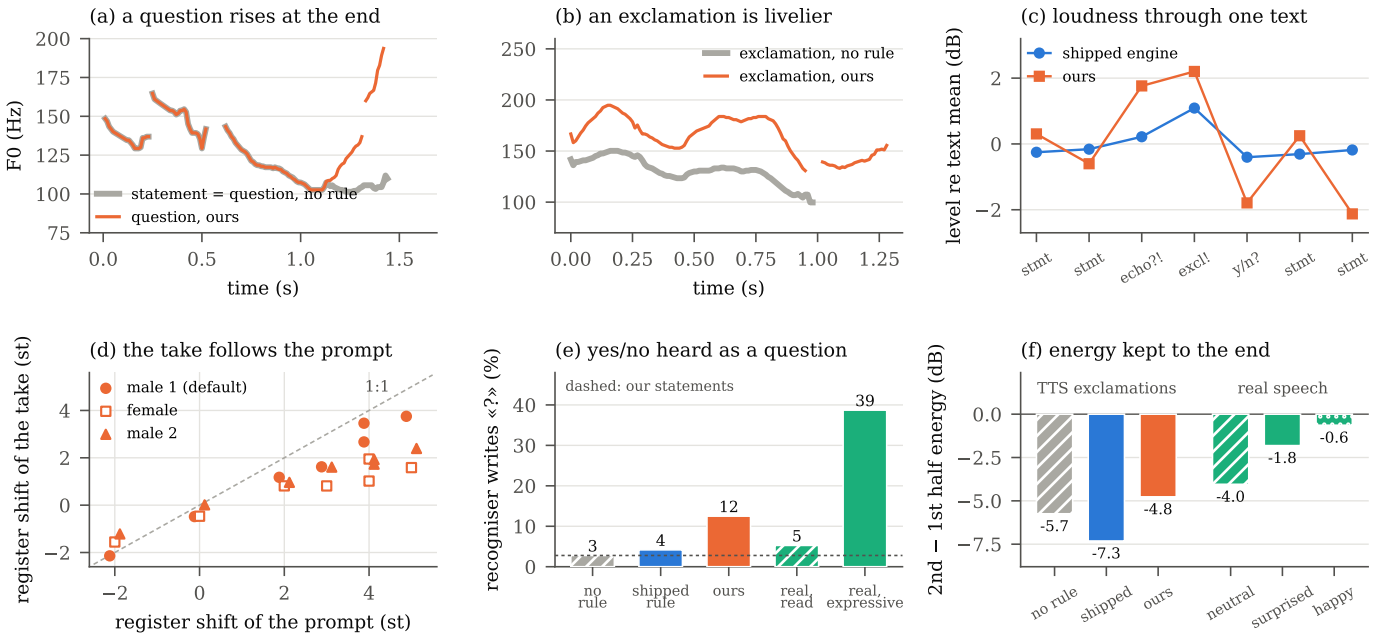

(a) a question rises at the end
F0 (Hz)
200
175
150
125
100
75
statement = question, no rule
question, ours
0.0
0.5
1.0
1.5
time (s)
(b) an exclamation is livelier
250
200
150
100
exclamation, no rule
exclamation, ours
0.00
0.25
0.50
0.75
1.00
1.25
time (s)
(c) loudness through one text
level re text mean (dB)
shipped engine
ours
2
0
−2
stmt
stmt
echo?!
excl!
y/n?
stmt
stmt
(d) the take follows the prompt
register shift of the take (st)
male 1 (default)
female
male 2
1:1
4
2
0
−2
−2
0
2
4
register shift of the prompt (st)
(e) yes/no heard as a question
recogniser writes «?» (%)
dashed: our statements
40
30
20
10
0
3
4
12
5
39
no rule
shipped rule
ours
real, read
real, expressive
(f) energy kept to the end
2nd − 1st half energy (dB)
TTS exclamations
real speech
0.0
−2.5
−5.0
−7.5
-5.7
-7.3
-4.8
-4.0
-1.8
-0.6
no rule
shipped
ours
neutral
surprised
happy